YuEEG / REV F / COMPACT SINGLE-SIDE SMD / STACKED AMOLED / SEPARATE POWER BOARD

01 / Electrode inputs CH1-CH4

File: inputs_1_4.kicad_sch

01 / Electrode inputs CH5-CH8

File: inputs_5_8.kicad_sch

02 / ADS1299 digital, bias, reference and decoupling

File: afe_core.kicad_sch

03 / ESP32-S3 wireless acquisition controller

File: mcu_radio.kicad_sch

04 / Battery input and independent analog-digital supplies

File: power.kicad_sch

05 / Bipolar analog supply

File: power_bipolar.kicad_sch

06 / Mainboard isolated DATA port and battery interlock

File: battery_charger.kicad_sch

07 / Raw 2S cell protection and fail-safe acquisition switch

File: battery_protection.kicad_sch

08 / Cell balancing with raw-domain UV qualification

File: battery_balance.kicad_sch

09 / Isolated USB DATA only - no board power path

File: isolated_usb.kicad_sch

10 / Protected 2S charge harness - secondary only

File: usb_system_power.kicad_sch

12 / AMOLED 1.91 inch RM67162 - DWO DO0200PFS12

File: amoled_ui.kicad_sch

13 / SDMMC backup, status and low-profile controls

File: storage_status.kicad_sch

14 / DATA port passive protection - CHARGE port moved offboard

File: dual_usb_ports.kicad_sch

15 / Wet electrode harness - compact single-row connector

File: electrode_header.kicad_sch

Engineering review. Screen, storage, USB, battery and patient isolation require physical validation.
Battery-only acquisition; detached isolated charger. Follow the Rev F assembly notes.

DUAL USB / AMOLED / SD BACKUP - ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /
File: YuEEG_8CH_RevF.kicad_sch

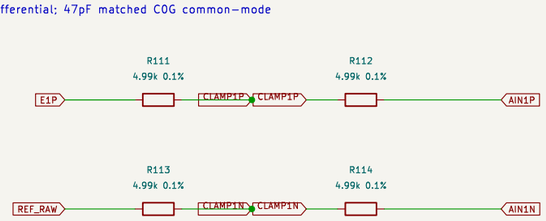
Title: YuEEG 8CH - wireless EEG acquisition

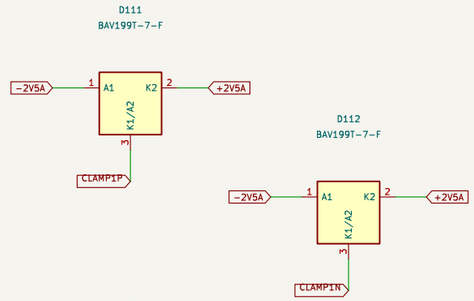
Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 1/16

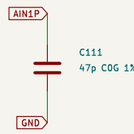
01 / ELECTRODE INPUTS CH1–CH4

Referential EEG: one SIGNAL terminal per channel. Shared REF –> all N input networks. BIAS is a separate driven electrode.

CH1 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode



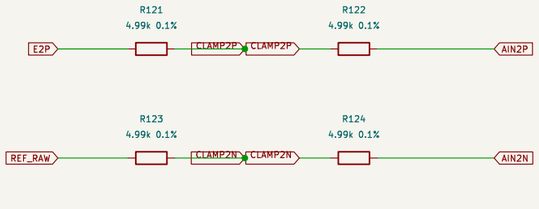


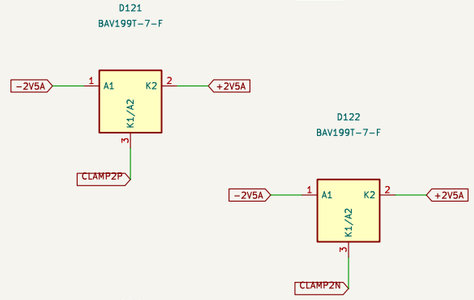


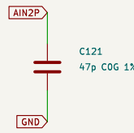




CH2 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode



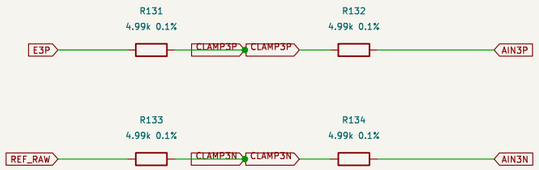


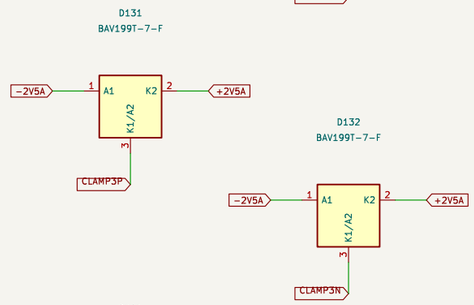


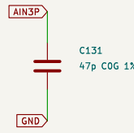


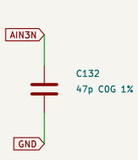


CH3 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode



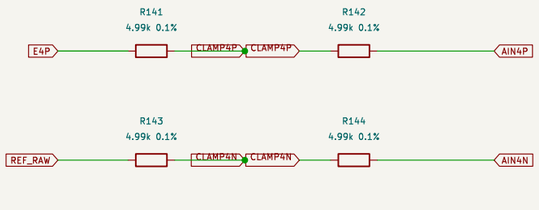


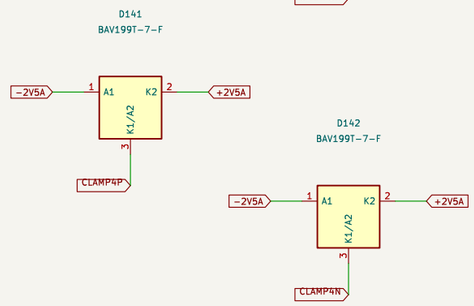


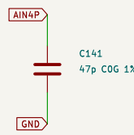




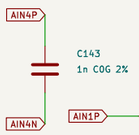
CH4 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common–mode

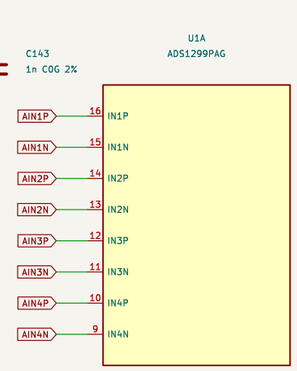












Front–end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/–2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current–limited through the N–side resistors.
No electrode is connected directly to board GND.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /01 / Electrode inputs CH1–CH4/
File: inputs_1_4.kicad_sch

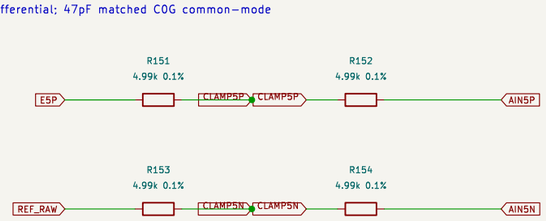
Title: 01 / Electrode inputs CH1–CH4

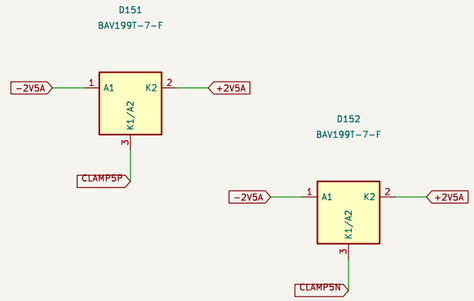
Size: A3	Date: 2026–10–27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 2/16

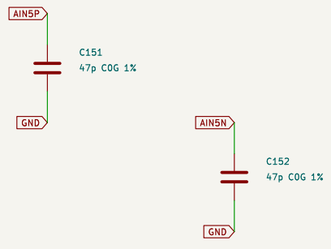
01 / ELECTRODE INPUTS CH5–CH8

Referential EEG: one SIGNAL terminal per channel. Shared REF -> all N input networks. BIAS is a separate driven electrode.

CH5 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common-mode

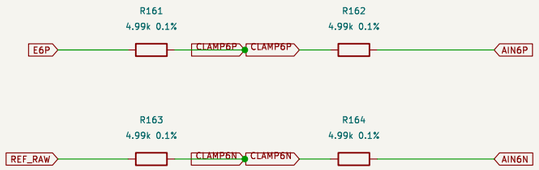


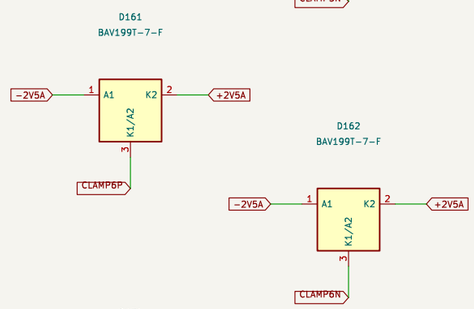


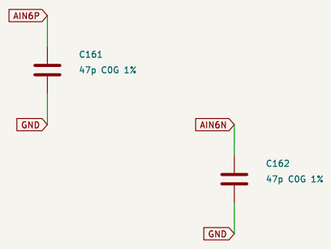




CH6 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common-mode

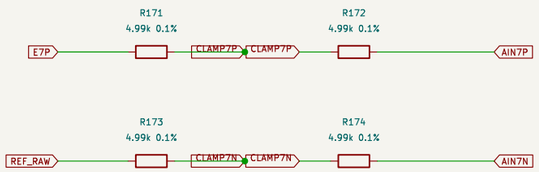


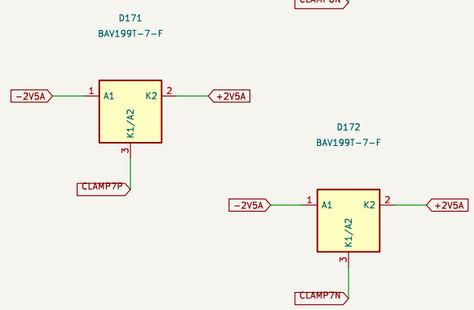


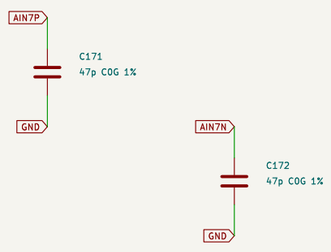




CH7 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common-mode

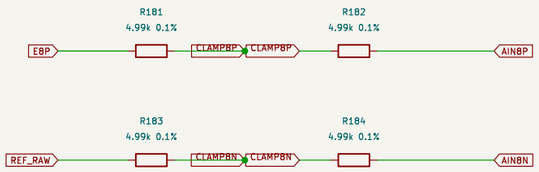


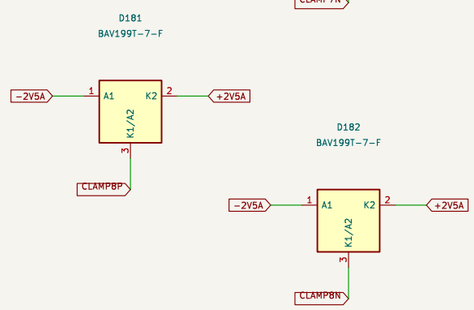


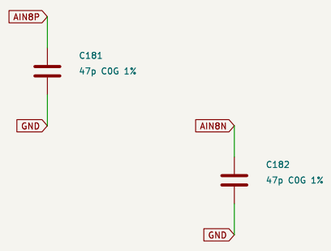


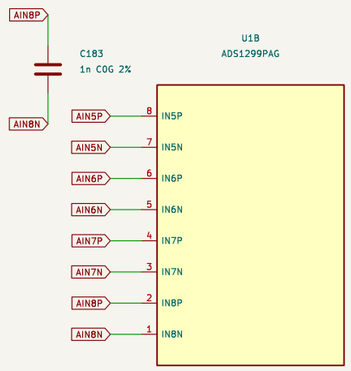


CH8 | 2 x 4.99k per lead; 1nF differential; 47pF matched COG common-mode









Front-end RC is RF/alias prefiltering, not the 0.5–80Hz digital bandpass.
BAV199 steers to +/−2.5V; rail steering, NOT certified patient protection or a 0.3V hard clamp.
Reference leads are also current-limited through the N-side resistors.
No electrode is connected directly to board GND.

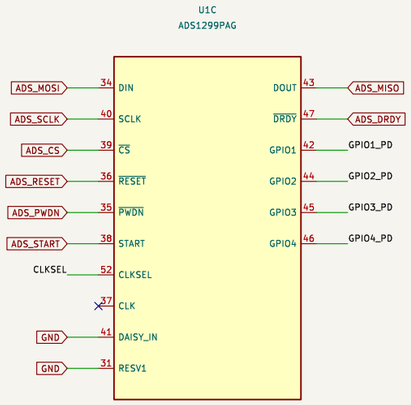
DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG

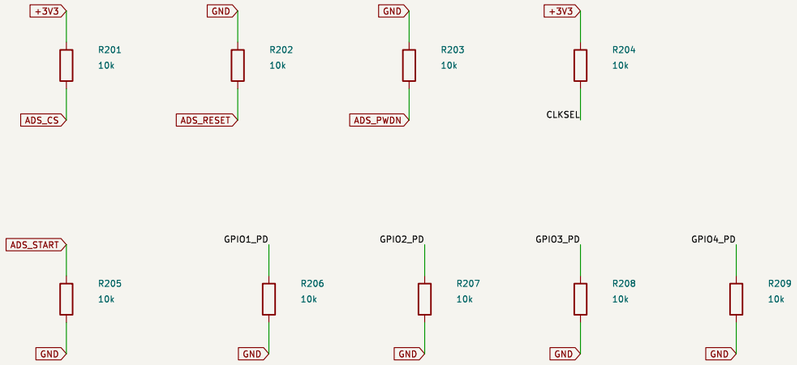
Sheet: /01 / Electrode inputs CH5–CH8/
File: inputs_5_8.kicad_sch

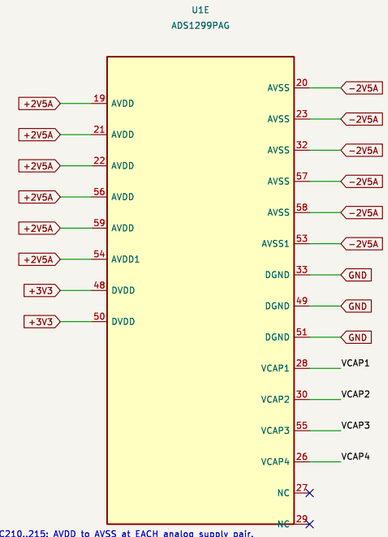
Title: 01 / Electrode inputs CH5–CH8

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 3/16

02 / ADS1299 DIGITAL, BIAS, REFERENCE AND DECOUPLING

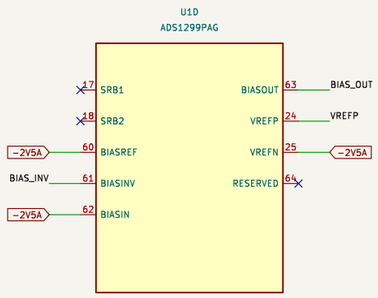


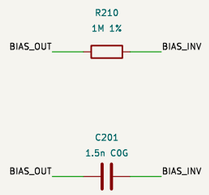




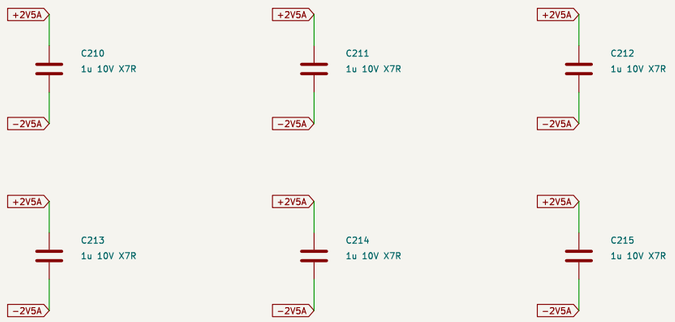
Internal 2.048MHz clock: CLKSEL pulled high; CLK output disabled.
Use SPI MODE1. DRDY -> ISR notification -> SPI/DMA task -> packet FIFO.
RESET/PWDN default LOW. Release PWDN after rails settle; hold RESET.
Wait 10s initially; verify VCAP1−AVSS >1.1V BEFORE releasing RESET.

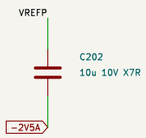
C210..215: AVDD to AVSS at EACH analog supply pair.
C216/217: one at EACH digital supply pin.
Keep VCAP and VREF loops short.





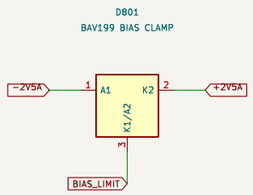


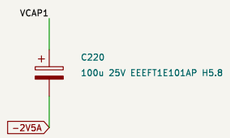


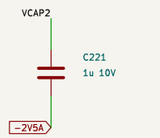


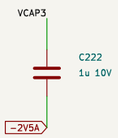


BIASREF_INT=1 -> internal near−0V midpoint. SRB1/SRB2 unused.
Use CHnSET=0x60 and MISC1=0x00 (external N inputs).
Start with BIAS_SENSP=0xFF, BIAS_SENSN=0x00; validate loop stability.
BIAS feedback 1M || 1.5nF is a TI example, not a guaranteed stable loop.
Lead−off excitation disabled in baseline recording; enable only in a separate test mode.

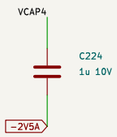












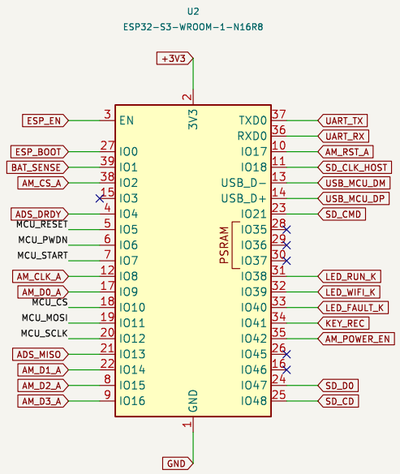
DUAL USB / AMOLED / SD BACKUP — ENGINEERING VALIDATION REQUIRED
YuEEG

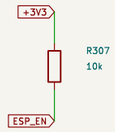
Sheet: /02 / ADS1299 digital, bias, reference and decoupling/
File: afe_core.kicad_sch

Title: 02 / ADS1299 digital, bias, reference and decoupling

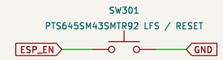
Size: A3	Date: 2026−10−27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 4/16

03 / ESP32-S3 WIRELESS ACQUISITION CONTROLLER











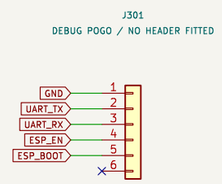




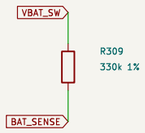


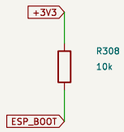






J301 is 3.3V UART, not RS-232.
No USB or charging power path is fitted.
Disconnect ALL electrodes before attaching
UART, debugger, scope or external instruments.



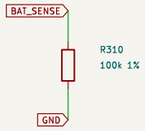


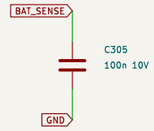












VBAT/4.3 -> GPIO1 ADC1. Max 8.4V -> 1.95V.
Calibrate ADC and stop acquisition below 6.4V.
Pack protection is external; this divider is NOT a BMS.

GPIO map: IO4 DRDY / IO5 RESET / IO6 PWDN / IO7 START / IO10 CS / IO11 MOSI / IO12 SCLK / IO13 MISO.
Boot strap pins IO3, IO45, IO46 are unused. Choose N8 (no PSRAM), not an arbitrary module variant.
Transmit int24 samples + sequence counter + sample-rate/gain metadata; target 500SPS, 24x.
Do not print volts from the interrupt. This hardware requires new pin mapping and firmware integration.
RF module antenna at PCB edge; preserve Espressif antenna keep-out on all layers.

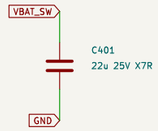
DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /03 / ESP32-S3 wireless acquisition controller/
File: mcu_radio.kicad_sch
Title: 03 / ESP32-S3 wireless acquisition controller
Size: A3 | Date: 2026-10-27 | **Rev: F**
KiCad E.D.A. 9.0.5 | Id: 5/16

04 / BATTERY INPUT AND INDEPENDENT ANALOG–DIGITAL SUPPLIES

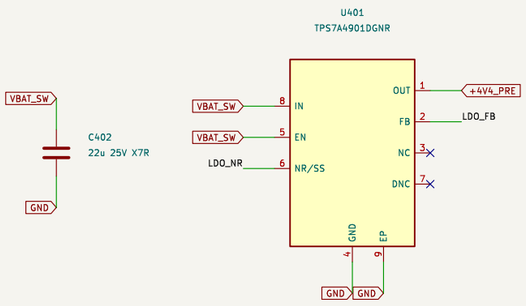


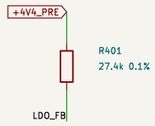


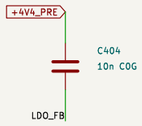


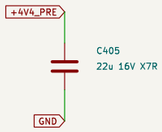


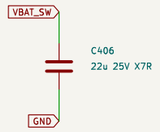
Onboard 2S charger/protection/balancer: see battery–management sheets.
J401 now has B+, MID, B–; connect a matched 2S pack plus a separate insulated NTC.
Isolated USB may supply acquisition via delayed SYS path. POWER OFF for charging. System isolation validation pending.

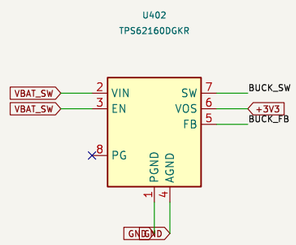




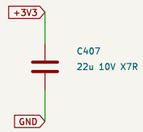


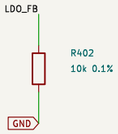


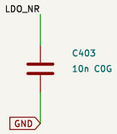


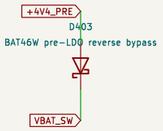


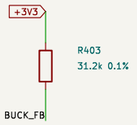


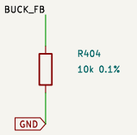












Pre–regulator (NOT ADS AVDD): 1.185 x (1 + 27.4k/10k) = 4.432V nominal.
22uF nominal input/output; >=10uF effective after DC bias.
Feeds TPS73225 and TPS60403 on bipolar supply sheet. LDO EP to GND. Keep digital switching return away from AFE.















Digital: 0.8 x (1 + 31.2k/10k) = 3.296V nominal.
VOS senses output AFTER inductor. 1A rated converter.
Minimize VIN–cap/SW/PGND hot loop; validate Wi–Fi transients.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED

YuEEG

Sheet: /04 / Battery input and independent analog–digital supplies/
File: power.kicad_sch

Title: 04 / Battery input and independent analog–digital supplies

Size: A3	Date: 2026–10–27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 6/16

BIPOLAR ANALOG SUPPLY: +2.5V / −2.5V; independent 3.3V digital ground reference

RESET/PWDN default LOW. After rails settle: PWDN HIGH, keep RESET LOW.
Wait >=tPOR AND VCAP1−AVSS >1.1V; use 10s initial bench delay.
100uF / internal 9.8k means seconds, NOT 200ms. Verify with scope.
Then RESET HIGH; wait >=18 CLK; SDATAC/configure/read back/start.
No automatic analog−rail/VCAP monitor is fitted; qualify timing on bench.

+4V4_PRE
C410
1u 10V X7R
GND

U403
TPS73225DBVR
+4V4_PRE
+4V4_PRE
IN
EN
GND
OUT
NR
+2V5A
POS_NR
GND

+2V5A
C412
10u 10V X7R
GND

+2V5A
R410
4.99k 1%
GND

POS_NR
C411
10n COG
GND

C414
2.2u 10V X7R
CFLY_PLUS
CFLY_MINUS

NEG_RAW
D402
BAT46W reverse−current bypass
−2V5A

+4V4_PRE
C413
2.2u 10V X7R
GND

U404
TPS60403DBVR
+4V4_PRE
IN
GND
OUT
CFLY+
CFLY−
NEG_RAW
CFLY_PLUS
CFLY_MINUS
GND

GND
C415
10u 10V X7R
NEG_RAW

U405
TPS72325DBVR
NEG_RAW
+3V3
IN
EN
GND
OUT
NR
−2V5A
NEG_NR
GND

GND
C417
10u 10V X7R
−2V5A

GND
R411
4.99k 1%
−2V5A

NEG_NR
C416
10n COG
GND

GND
C418
1u 10V X7R
NEG_RAW

TP405
+2V5A
+2V5A

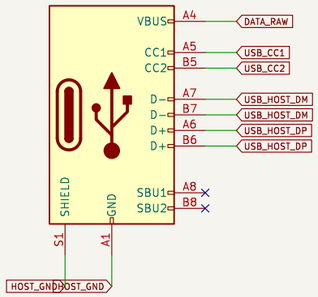
TP406
−2V5A
−2V5A

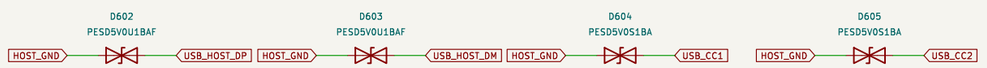
TP407
NEG_RAW
NEG_RAW

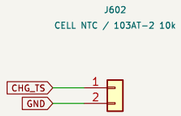
4.43V pre−regulator −> positive LDO: 250kHz inverter −> negative low−noise LDO.
Keep CFLY/input/output capacitor loops at U404. Do not route switching nodes into the AFE.
TPS723 EN is +3V3 (not +4V4_PRE); 10nF NR to GND. D402 bypasses reverse current at power−down.
4.99k rail bleeders provide ~0.5mA load/discharge; they do not absorb arbitrary electrode faults.

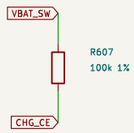
DUAL USB / AMOLED / SD BACKUP − ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /05 / Bipolar analog supply/
File: power_bipolar.kicad_sch
Title: 05 / Bipolar analog supply and power−up requirements

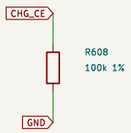
Size: A3	Date: 2026−10−27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 7/16











DATA/FLASH supplies ADuM upstream only. Mainboard acquisition uses protected battery.
J602 is the pack 10kohm NTC (103AT-2); this signal and CHG_CE reach the isolated charger through J1202.
The primary CHARGE USB port, source-current gate, isolated supply and BQ25886 are all on the power daughterboard.
No charger SYS output connects to mainboard ACQ_INPUT. Physically unplug the electrode harness before charging.

BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

YuEEG

Sheet: /06 / Mainboard isolated DATA port and battery interlock/
File: battery_charger.kicad_sch

Title: 06 / Mainboard isolated DATA port and battery interlock

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 8/16

RAW CELL HARNESS: B+ / MID / B− ; NEVER SHORT RAW B− TO SYSTEM GND

J401
2S RAW: B+ / MID / B−
RAW_BPLUS
RAW_MID
RAW_BMINUS

F601
1A fast fuse / 32V
RAW_BPLUS
PACK_POS

U602
S-8252AAY-M6T1U
PROT_VDD
PROT_VC
PROT_VM
VDD
VC
VM
DO
CO
VSS
DSG_GATE
CHARGE_GATE
RAW_BMINUS

U604
TPS3700DDCR RUN WINDOW
PACK_POS
RUN_UV
RUN_OV
VDD
INA+
INB−
OUTA
OUTB
GND
RUN_BASE
RUN_BASE
GND

C640
100n 16V
PACK_POS
GND

R620
470R 1%
RAW_BPLUS
PROT_VDD

R621
470R 1%
RAW_MID
PROT_VC

C620
100n 16V
PROT_VDD
RAW_BMINUS

C621
100n 16V
PROT_VC
RAW_BMINUS

R622
2.00k 1%
GND
PROT_VM

R640
1.58M 0.1%
PACK_POS
RUN_UV

R641
100k 0.1%
RUN_UV
GND

R642
2.05M 0.1%
PACK_POS
RUN_OV

R643
100k 0.1%
RUN_OV
GND

Q601
CSD17308Q3 DSG
DSG_SOURCE
DSG_SOURCE
DSG_SOURCE
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
DSG_GATE

Q602
CSD17308Q3 CHG
GND
GND
GND
S1
S2
S3
D5-8_EP
G
BMS_DRAINS
CHARGE_GATE

R623
0.010R 1% 0.5W
RAW_BMINUS
DSG_SOURCE

C641
10n C0G
RUN_UV
GND

R644
62k 1%
PACK_POS
RUN_BASE

R645
10k
RUN_BASE
RUN_NPN_BASE

R646
1M
RUN_NPN_BASE
GND

Q605
MMBT3904 ENABLE
RUN_PULLDOWN
RUN_NPN_BASE
GND

D606
BZT52C8V2S
DSG_GATE
DSG_SOURCE

D607
BZT52C8V2S
CHARGE_GATE
GND

R650
100k
RUN_GATE
RUN_PULLDOWN

Q603
AO3401A BACK-TO-BACK
PACK_POS
RUN_GATE
RUN_SOURCES

Q604
AO3401A BACK-TO-BACK
ACQ_INPUT
RUN_GATE
RUN_SOURCES

R649
1M
RUN_SOURCES
RUN_GATE

C642
10n 25V
RUN_GATE
RUN_SOURCES

R651
47k
ACQ_INPUT
GND

#FLG610
PWR_FLAG
RAW_BPLUS

#FLG611
PWR_FLAG
RAW_BMINUS

#FLG612
PWR_FLAG
PACK_POS

#FLG613
PWR_FLAG
PROT_VDD

Primary protection per cell: OV 4.25V / UV 3.00V; inhibit OV-cell recharge. Charge/discharge overcurrent use VM and actual series resistance.
Q601/Q602 are common-DRAIN NMOS. Q603/Q604 are common-SOURCE PMOS: no body-diode bypass when OFF.
RUN typical falling cutoff ~6.63V, restart ~6.72V. OV cutoff ~8.6V. USB clamps RUN_BASE independently of MCU.
Raw B− returns only to protection/balance circuits. USB, charger, NTC and EEG use protected GND.
Hardware power-off is NOT galvanic or patient isolation. Remove electrodes before inserting USB.

DUAL USB / AMOLED / SD BACKUP — ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /07 / Raw 2S cell protection and fail-safe acquisition switch/
File: battery_protection.kicad_sch

Title: 07 / 2S protection and acquisition inhibit

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 9/16

CELL BALANCING: ONLY WITH VALID USB SOURCE AND RAW PACK ABOVE ~7.6V

RAW_BPLUS
R630
160R 1%
BAL_VC2

RAW_BPLUS
R633
100R 1%
BAL_VDD

BAL_VDD
C630
100n 16V
RAW_BMINUS

U603
BQ29209DRBR
BAL_VDD 7 VDD
BAL_VC2 1 VC2
BAL_VC1 2 VC1
BAL_VC1_CB 3 VC1_CB
BAL_ENABLE 6 CB_EN
OUT 8 BAL_OV_ALARM
CD 4 BAL_CD
GND 5
EP 9
RAW_BMINUS

RAW_MID
R631
260R 1%
BAL_VC1

BAL_VC2
C631
100n 16V
BAL_VC1

U605
TPS3700DDCR BAL UV
RAW_BPLUS 5 VDD
BAL_UV_SENSE 3 INA+
RAW_BMINUS 4 INB−
OUTA 1 RAW_OK
OUTB 6
GND 2
RAW_BMINUS

RAW_MID
R632
100R 1%
BAL_VC1_CB

BAL_VC1
C632
100n 16V
RAW_BMINUS

RAW_BPLUS
R660
1.81M 0.1%
BAL_UV_SENSE

BAL_UV_SENSE
R661
100k 0.1%
RAW_BMINUS

RAW_BPLUS
R662
1M
RAW_OK

RAW_BPLUS
C660
100n 16V
RAW_BMINUS

BAL_CD
C633
330n 16V
RAW_BMINUS

BAL_VDD
R634
330k
BAL_ENABLE

USB5
R663
2.2k
BAL_OPTO_A

U606
LTV−356T VALID USB
BAL_OPTO_A 1
CHG_PG 2
4 BAL_ENABLE
3 BAL_AND_SOURCE

Q607
2N7002 RAW UV GATE
BAL_AND_SOURCE
RAW_OK 1
3
2
RAW_BMINUS

TP601
RAW_BPLUS
RAW_BPLUS

TP602
RAW_MID
RAW_MID

TP603
RAW_BMINUS
RAW_BMINUS

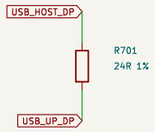
TP604
BAL_OV_ALARM
BAL_OV_ALARM

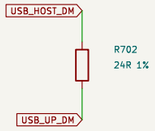
#FLG620
PWR_FLAG
BAL_VDD

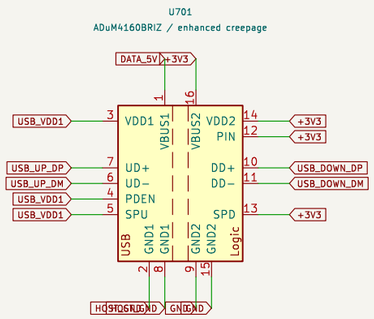
TP605
CHG_PG
CHG_PG

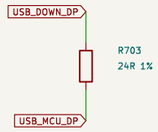
BQ29209 internal balancing: TI Fig9 resistor network, about 10mA; matched cells required. Not a cell equalizer for mismatched/old cells.
VC2 capacitor returns to VC1; VC1 capacitor returns to RAW B−. Do not return raw-domain nodes to system GND.
CB_EN active LOW: 330k pull-up disables by default. Optocoupler needs /PG LOW AND Q607 enabled by raw pack UV comparator.
Raw UV typical rising ~7.64V/falling ~7.53V; prevents balancing depleted cells. Optocoupler is domain translation, NOT patient isolation.
OUT is an alarm test point only: primary charge/discharge cutoff is independently provided by S−8252AAY.
J401 sequence and external NTC wiring must be verified before cells are plugged in. No standalone charger firmware is needed.

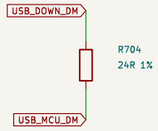
DUAL USB / AMOLED / SD BACKUP − ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /08 / Cell balancing with raw−domain UV qualification/
File: battery_balance.kicad_sch
Title: 08 / Cell balancing with raw−domain UV qualification
Size: A3 | Date: 2026−10−27 | **Rev: F**
KiCad E.D.A. 9.0.5 | Id: 10/16

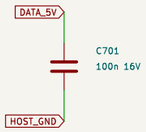


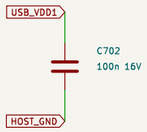




















ADuM4160 primary belongs only to DATA HOST_GND; the isolated charger primary is a separate board and ground.
Secondary uses the battery-powered system +3V3/GND. USB cannot start acquisition without a battery.
Preserve 8mm copper separation, including all layers, retainers and user-accessible cables.

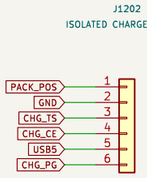
BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

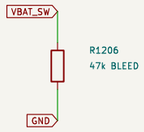
YuEEG

Sheet: /09 / Isolated USB DATA only — no board power path/
File: isolated_usb.kicad_sch

Title: 09 / Isolated USB DATA only — no board power path

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 11/16





Only ISOLATED SECONDARY signals reach this connector. No HOST ground is carried in this harness.
1 PACK_POS, 2 GND, 3 pack NTC, 4 CHG_CE, 5 isolated USB5, 6 CHG_PG. Match power-board J1204 pin-for-pin.
R607/R608 hold CE high during acquisition. Daughterboard R1205 defaults CE high if the harness/CE wire is open.
POWER switch now controls the battery MOSFET gate logic; it does not carry battery load current.
R1206 discharges switched VBAT after OFF; charging begins only after CE falls below its threshold.
Do not charge while connected to a person: physically unplug every electrode, including REF and BIAS.
Use a keyed insulated GH6 harness, rated >=1A, target length <=50mm. Validate pin order before connecting cells.

BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

YuEEG

Sheet: /10 / Protected 2S charge harness — secondary only/
File: usb_system_power.kicad_sch

Title: 10 / Protected 2S charge harness — secondary only

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 12/16

12 / AMOLED 1.91 INCH RM67162 – DWO DO0200PFS12

U1001
TPS22918DBVR AMOLED POWER
+3V3
AM_POWER_EN
AM_CT
AM_RAW
AM_QOD
GND
C1003
100n 10V
U1002
TPS7A2028PDBVR
AM_RAW
AM_2V8
C1004
1u 10V
C1005
4.7u 10V
C1002
22u 10V X7R
C1008
100n 10V
U1004
SN74AXC8T245PWR / 3V3 -> 1V8
AM_1V8
C1009
100n 10V
U1003
TPS7A2018PDBVR
C1006
1u 10V
C1007
4.7u 10V
R1001
100k
C1001
1n 25V
R1003
1k
C1010
22u 10V X7R
C1011
1u 10V
C1012
100n 10V
Q1001
MMBT3904 OE ENABLE
AM_OE_N
AM_OE_BASE
R1004
100k
R1002
10k
R1006
10k
R1005
1M
J1001
AMOLED / FTSH-110-01-L-DV-K-P-TR
AM_IM1
R1031
47k
R1032
47k
R1033
47k
R1034
47k
R1035
47k
R1036
47k
R1037
47k
AM_CS_A
AM_CLK_A
AM_D0_A
AM_D1_A
AM_D2_A
AM_D3_A
AM_RST_A
R1021
22R
R1022
22R
R1023
22R
R1024
22R
R1025
22R
R1026
22R
R1027
22R
AM_CS_B
AM_CLK_B
AM_D0_B
AM_D1_B
AM_D2_B
AM_D3_B
AM_RST_B
AM_CS_P
AM_CLK_P
AM_D0_P
AM_D1_P
AM_D2_P
AM_D3_P
AM_RST_P

Lock RM67162 version; module R4=0R, R3=NC, IM1 high for QSPI. VBAT=AM_RAW; VCC=2.8V; IOVCC=1.8V.
Touch held in RESET; touch signals and TE unused. 18-core harness pin numbering must match vendor CN1.
Keep screen and harness in floating system domain. Driver starts at 10MHz; refresh limited to status regions.
Dedicated SPI3 QSPI; ADC uses SPI2; microSD uses separate 1-bit SDMMC. Supplies, translation and inrush require bench tests.

DUAL USB / AMOLED / SD BACKUP – ENGINEERING VALIDATION REQUIRED
YuEEG
Sheet: /12 / AMOLED 1.91 inch RM67162 – DWO DO0200PFS12/
File: amoled_ui.kicad_sch
Title: 12 / AMOLED 1.91 inch RM67162 – DWO DO0200PFS12
Size: A3 | Date: 2026-10-27 | **Rev: F**
KiCad E.D.A. 9.0.5 | Id: 13/16

13 / SDMMC BACKUP, STATUS AND LOW-PROFILE CONTROLS

+3V3

FB1101
BLM18PG121SN1D / 120R@100MHz

SD_3V3

J1101
HIROSE DM3AT-SF-PEJM5

SD_DAT2 — 1 — DAT2
SD_DAT3 — 2 — DAT3
SD_CMD — 3 — CMD
SD_3V3 — 4 — VDD
SD_CLK_CARD — 5 — CLK
GND — 6 — VSS
SD_D0 — 7 — DAT0
SD_DAT1 — 8 — DAT1
CD1 — 9 — SD_CD
CD2 — 10 — GND
SHIELD — 11 — GND

SD_3V3 — C1101 22u 10V X7R — GND
SD_3V3 — C1102 100n 10V — GND
SD_3V3 — C1103 1u 10V — GND

SD_CLK_HOST — R1101 22R — SD_CLK_CARD

D1101
USBLC6-2SC6
SD_3V3 — 5 VBUS
SD_CLK_CARD — 1 IO1 — 6 IO1 — SD_CLK_CARD
SD_CMD — 3 IO2 — 4 IO2 — SD_CMD
2 GND — GND

D1102
USBLC6-2SC6
SD_3V3 — 5 VBUS
SD_D0 — 1 IO1 — 6 IO1 — SD_D0
SD_CD — 3 IO2 — 4 IO2 — SD_CD
2 GND — GND

SD_3V3 — R1111 10k — SD_CMD
SD_3V3 — R1112 10k — SD_D0
SD_3V3 — R1113 10k — SD_DAT1
SD_3V3 — R1114 10k — SD_DAT2
SD_3V3 — R1115 10k — SD_DAT3
SD_3V3 — R1116 10k — SD_CD

#FLG1101
PWR_FLAG
SD_3V3

+3V3 — R1120 10k — KEY_REC

+3V3 — R840 2.2k — STATUS_POWER_A
LED801 POWER: GND — STATUS_POWER_A
+3V3 — R841 2.2k — STATUS_RUN_A
LED802 RUN: LED_RUN_K — STATUS_RUN_A
+3V3 — R842 2.2k — STATUS_WIFI_A
LED803 WIFI: LED_WIFI_K — STATUS_WIFI_A
+3V3 — R843 2.2k — STATUS_FAULT_A
LED804 FAULT: LED_FAULT_K — STATUS_FAULT_A

SW1001
PTS645SM43SMTR92 LFS / REC
KEY_REC — GND

KEY_REC — C1120 10n 10V — GND

H1001
AMOLED M2 / 8mm STANDOFF

H1002
AMOLED M2 / 8mm STANDOFF

H1003
AMOLED M2 / 8mm STANDOFF

H1004
AMOLED M2 / 8mm STANDOFF

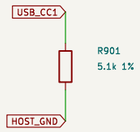
1-bit SDMMC: CMD and DAT0..3 each have 10k pull-ups, including unused DAT1/2/3. CLK has source series termination.
No SD, display or network IO in DRDY ISR. Separate recording ring and explicit drop counters; no lossless claim.
REC key toggles EEG start/stop after debounce. BOOT/RESET remain hardware buttons. All are 4.3mm low-profile metal dome switches.
AMOLED model is an assembly envelope above 6mm insulated standoffs; check cable contacts and mechanical stack with actual parts.

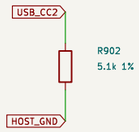
DUAL USB / AMOLED / SD BACKUP — ENGINEERING VALIDATION REQUIRED
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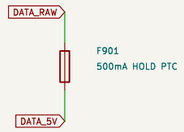
Sheet: /13 / SDMMC backup, status and low-profile controls/
File: storage_status.kicad_sch

Title: 13 / SDMMC backup, status and low-profile controls

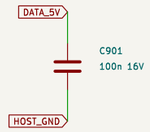
Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 14/16













Two Type-C ports remain in the complete system: DATA/FLASH on mainboard; CHARGE on the detachable power board.
Never connect either charger primary ground or DATA HOST_GND directly to system GND.

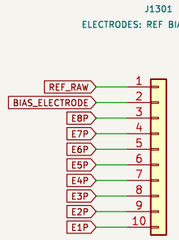
BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

YuEEG

Sheet: /14 / DATA port passive protection — CHARGE port moved offboard/
File: dual_usb_ports.kicad_sch

Title: 14 / DATA port passive protection — CHARGE port moved offboard

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 15/16



One 2.54mm-pitch connector replaces nine separate header bodies.
Pin 1 REF; pin 2 BIAS; pins 3..10 CH8..CH1. Labels follow the physical order.
No scalp electrode connects directly to system GND.
Short raw-electrode stubs enter the matched protection/filter bank, then ADS1299 inputs.
AMOLED stacks ABOVE the mainboard on 8mm insulated standoffs, to the right of this connector.
Keep the left plug-in zone and the ESP antenna clear. All mainboard SMD parts are on TOP.
Before charging, physically unplug the complete electrode harness, including REF and BIAS.
Matched isolated power daughterboard is detachable; no USB SYS path powers acquisition.

BATTERY ONLY — NOT VALIDATED FOR HUMAN CONNECTION

YuEEG

Sheet: /15 / Wet electrode harness — compact single-row connector/
File: electrode_header.kicad_sch

Title: 15 / Wet electrode harness — compact single-row connector

Size: A3	Date: 2026-10-27	**Rev: F**
KiCad E.D.A. 9.0.5		Id: 16/16